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Small Write Up

During this project, I implemented a graph-based system of roads and towns to connect with each other through routes. Each Town is represented by a vertex, and each Road is represented by a weighted (distance) edge between the towns. The Graph class is a structure that manages the adjacent lists and supports operations such as adding/removing vertices and edges, checking for the right connections, then computing the shortest paths only by using the Dijkstra algorithm.

The TownGraphManager is provided for a simple interface for the classes of Town to directly flow easier with the names, the retrieve paths, and the load graph data from a file. Overall, the project has done a great job in having me demonstrate core graph concepts, objects, and shortest path algorithm implementation.

The tests I ran were the Graph_STUDENT_Test, Road_STUDENT_Test, and the Town_STUDENT_Test, which were crucial to finding any bugs that could be hidden deep inside the code. These were made to find out in each class how I can fix up the code to make it flow better and have a much easier time finding errors.

JavaFX Outcomes

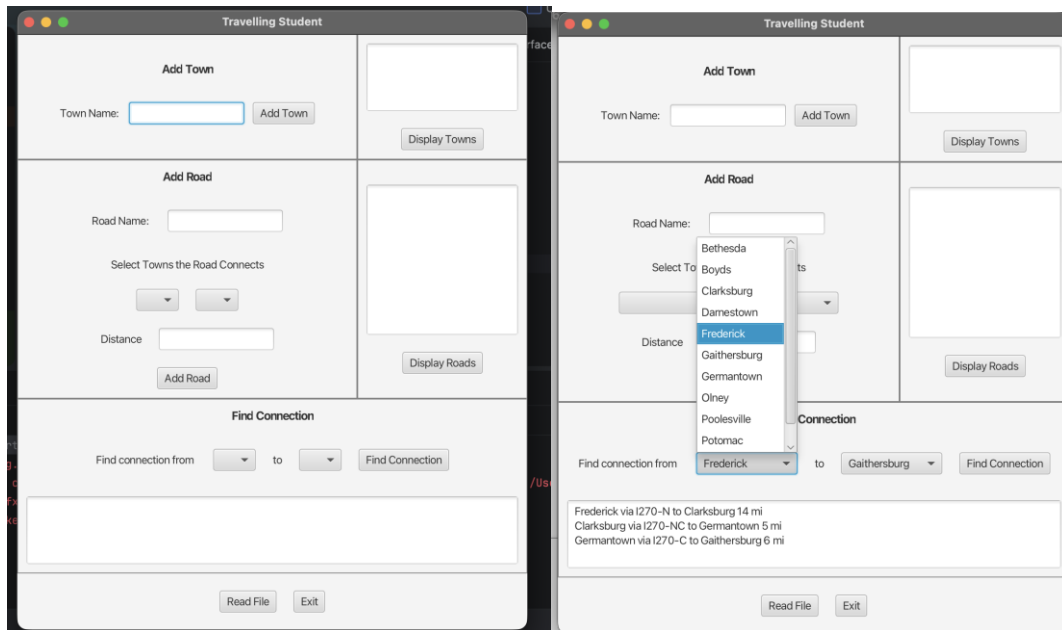
The image displays four screenshots of a macOS-style application titled "Travelling Student".

Top Left Screenshot: Shows the initial state. The "Add Town" section has a "Town Name" input field and an "Add Town" button. The "Add Road" section has a "Road Name" input field, two dropdown menus for "Select Towns the Road Connects", a "Distance" input field, and an "Add Road" button. The "Find Connection" section has two dropdown menus for "Find connection from" (set to "Frederick") and "to" (set to "Gaithersburg"), a "Find Connection" button, and a text area displaying the route: "Frederick via I270-N to Clarksburg 14 mi", "Clarksburg via I270-NC to Germantown 5 mi", and "Germantown via I270-C to Gaithersburg 6 mi". There are "Read File" and "Exit" buttons at the bottom.

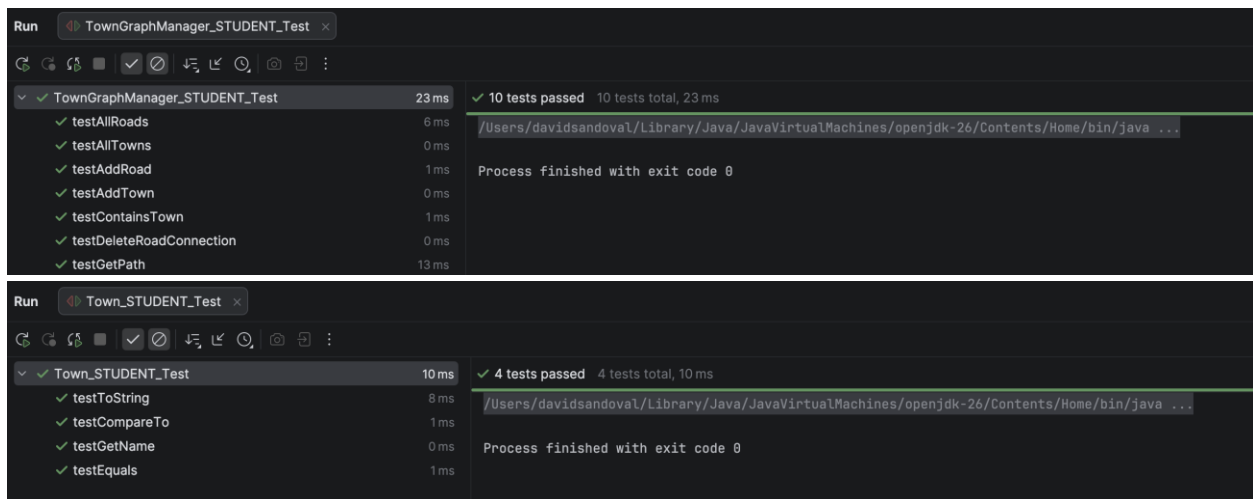
Top Right Screenshot: Shows the application after adding a new town and road. The "Add Town" section now has a "Display Towns" button. The "Add Road" section now has a "Display Roads" button. The "Find Connection" section now shows a different route: "Bethesda via MD190E to Potomac 7 mi", "Potomac via MD190W to Darnestown 8 mi", "Darnestown via MD118W to Germantown 6 mi", and "Germantown via I270-NC to Clarksburg 5 mi".

Bottom Left Screenshot: Shows an error dialog box titled "Error" with the message "Road name cannot be blank". The dialog has an "OK" button. The background application is dimmed.

Bottom Right Screenshot: Shows another error dialog box titled "Error" with the message "Town name cannot be empty". The dialog has an "OK" button. The background application is dimmed.



Screenshots of Tests



The screenshot displays the Run console of an IDE, showing the execution of four test classes. Each class has a list of tests with their individual execution times and a summary of the total tests passed.

Run Road_STUDENT_Test

Test Name	Time (ms)
testToString	6 ms
testCompareTo	0 ms
testGetName	2 ms
testContains	1 ms
testEquals	0 ms
testGetWeight	0 ms

6 tests passed 6 tests total, 9 ms

Process finished with exit code 0

Run Graph_STUDENT_Test

Test Name	Time (ms)
testContainsVertex	8 ms
testAddEdge	2 ms
testContainsEdge	0 ms
testVertexSet	1 ms
testGetEdge	1 ms
testRemoveVertex	1 ms
testRemoveEdge	0 ms
testAddVertex	0 ms
testEdgeSet	1 ms

9 tests passed 9 tests total, 14 ms

Process finished with exit code 0

Run TownGraphTest

Test Name	Time (ms)
testTown1ToTown_10	19 ms
testTown_1ToTown_11	1 ms
testContainsVertex	0 ms
testAddEdge	0 ms
testContainsEdge	0 ms
testVertexSet	0 ms
testGetEdge	0 ms
testRemoveVertex	1 ms
testRemoveEdge	0 ms
testAddVertex	1 ms
testEdgeSet	0 ms
testEdgesOf	0 ms
testTown_4ToTown_11	0 ms

13 tests passed 13 tests total, 22 ms

Process finished with exit code 0

Run TownGraphManagerTest

Test Name	Time (ms)
testAllRoads	26 ms
testAllTowns	1 ms
testGetPathA	23 ms
testGetPathB	1 ms
testAddRoad	0 ms
testAddTown	0 ms
testContainsTown	1 ms
testDeleteRoadConnection	1 ms
testGetPath	1 ms
testGetRoad	1 ms
testContainsRoadConnection	1 ms
testDisjointGraph	0 ms
testDeleteTown	2 ms

13 tests passed 13 tests total, 58 ms

Process finished with exit code 0

The image displays three sequential screenshots of an IDE's Run console, showing the execution of different test classes. Each screenshot includes a toolbar with icons for running, debugging, and other actions, followed by a table of test results and a summary of the overall test status.

Run TownGraphManager_GFA_Test

Test Name	Duration
testAddRoad	24 ms

✓ 1 test passed 1 test total, 24 ms

Process finished with exit code 0

Run GraphTest

Test Name	Duration
testTown1ToTown_10	22 ms
testTown_1ToTown_11	1 ms
testContainsVertex	0 ms
testAddEdge	0 ms
testContainsEdge	1 ms
testVertexSet	0 ms
testGetEdge	0 ms
testRemoveVertex	1 ms
testRemoveEdge	0 ms
testAddVertex	1 ms
testEdgeSet	0 ms
testEdgesOf	1 ms
testTown_4ToTown_11	1 ms

✓ 13 tests passed 13 tests total, 28 ms

Process finished with exit code 0

Run Graph_GFA_Test

Test Name	Duration
testGetEdge	13 ms

✓ 1 test passed 1 test total, 13 ms

Process finished with exit code 0